

- **Problem Statement ID - 1**
- **Problem Statement Title - Autonomous AI Agent for Real-Time Building Energy Optimization**
- **Theme - Clean & Green Technology / Smart Automation**
- **PS Category - Software**
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EcoLoop — LLM-Driven Autonomous HVAC Control Agent

- A closed-loop AI agent that replaces static HVAC schedules with real-time, physics-grounded decisions.
- An autonomous agent pairs EnergyPlus building-physics simulation with a self-hosted Qwen2.5-7B LLM, which sets cooling/heating setpoints every simulated hour based on live zone temperature, outdoor conditions, and thermal trend.
- Traditional BMS use rigid fixed schedules that cannot adapt to real-time weather or occupancy shifts; our agent continuously reasons over live sensor data and self-corrects, discovering the building's true efficient operating envelope instead of guessing at design time.
- Combines native LLM tool-calling (zero fine-tuning) with day/night memory partitioning and rejection-as-feedback — turning hardware safety limits into a live, self-correcting signal, with every decision and model version fully auditable.

TECHNICAL APPROACH

- Technologies: EnergyPlus 26.1 + PyEnergyPlus API (simulation engine), Qwen2.5-7B-Instruct served via Ollama with native tool-calling, Python orchestration layer, ngrok tunnel for remote LLM access, pandas-based savings dashboard.
- Methodology: An Interval Governor triggers exactly one LLM inference per simulated hour; proposed setpoints pass through a rule-based Safety Net (physical + occupancy comfort bounds) before reaching EnergyPlus actuators; rejected actions feed back into a day/night memory buffer so the agent self-corrects, validated against a static baseline run for direct kWh comparison.

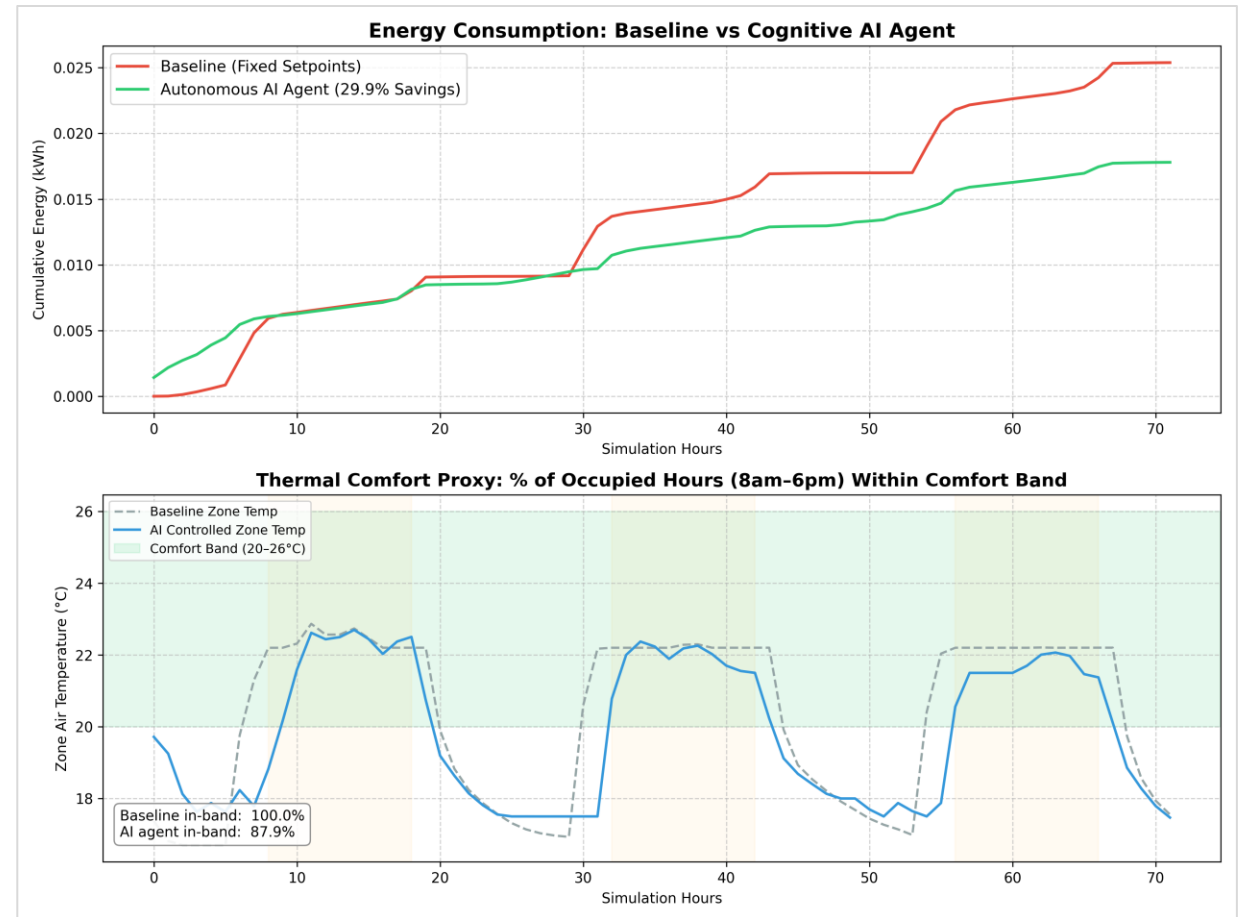
FEASIBILITY AND VIABILITY

- Feasibility: Built entirely on open-source, self-hostable components (EnergyPlus, Ollama, Qwen2.5) — no proprietary APIs or licensing cost, deployable on any building with a BMS-exposed setpoint interface.
- Challenges: LLM inference latency over network tunnels; risk of the model proposing physically invalid setpoints; balancing energy savings against occupant comfort.
- Mitigation: a bounded memory context caps prompt/latency growth; a deterministic safety-net layer hard-rejects invalid actions and feeds structured correction text back to the model; occupancy-aware comfort bands are enforced in code regardless of AI output.

ARTIFACTS

Verified Results — 72-hr Simulation (3 Day/Night Cycles)

- Baseline HVAC energy: 91.39 kWh
- Autonomous AI-agent energy: 64.09 kWh
- **Net savings: 27.30 kWh (29.87%)**
- Comfort in-band (20–26° C, occupied hrs): Baseline 100% | AI 87.9%
- Fail-static confirmed live: ngrok drop at Hour 0 → agent held last-known-good setpoints, no crash
- Full code, architecture doc & dashboard included in repo



RESEARCH AND REFERENCES

References:

- EnergyPlus Engineering Reference & PyEnergyPlus API docs (energyplus.net)
- Qwen2.5 model card (Ollama library)
- ASHRAE Standard 55 - Thermal Environmental Conditions for Human Occupancy
- Ollama tool-calling API documentation.